

Question Paper

Exam Date & Time: 28-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIFTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Satellite Communication [ECE_3126]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Provide the comparison between low earth orbit, medium earth orbit and geosynchronous Earth orbit satellites. (4)
- A) Course Outcome: "CO1" / Cognitive Level: "Application"
- B) Establish the need for Satellite Communication. (3)
- Course Outcome: "CO1" / Cognitive Level: "Application"
- C) Describe the following terms with respect to Satellite Communication (3)
- a. Subsatellite path
- b. Perigee
- c. Prograde orbit
- Course Outcome: "CO2" / Cognitive Level: "Understand"
- 2) With neat diagrams, explain the commonly employed techniques for satellite attitude control (stabilization techniques). Tabulate the differences between them. (5)
- A) Course Outcome: "CO2" / Cognitive Level: "Application"
- B) i) The apogee and perigee distances of a satellite orbiting in an elliptical orbit are respectively 45000 km and 7000 km. Calculate the following: (5)
- a. Semi-major axis of the elliptical orbit
- b. Orbit eccentricity. Comment about the shape of the satellite orbit and whether the satellite can be a GEO satellite.
- ii) A satellite is orbiting the Earth at an altitude of 600 km above the Earth's surface. Given that the radius of the Earth is 6371 km, calculate the maximum line-of-sight distance between the satellite and a point on the Earth's surface (i.e., the distance from the satellite to the horizon). Assume the elevation angle is 0° (the horizon case).
- Course Outcome: "CO2" / Cognitive Level: "Application"
- 3) Establish the need for batteries on board the satellite. Provide a detailed note on the different type of batteries used. (4)
- A) Course Outcome: "CO3" / Cognitive Level: "Application"
- B) With a neat block diagram, explain the satellite tracking system (4)

Course Outcome: "CO3" / Cognitive Level: "Application"

- C) List all the subsystems of a typical satellite (2)

Course Outcome: "CO3" / Cognitive Level: "Understand"

- 4) Provide a detailed note on the following related to the satellite communication link (5)
 (I). Free-space loss
 A) (II). Gaseous absorption

Course Outcome: "CO4" / Cognitive Level: "Application"

- B) (I). Discuss in detail the CDMA technique used in satellite communication (5)
 (II). List the advantages of TDMA over FDMA

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5) Outline and summarize any five applications of weather forecasting satellites in detail (5)

- A) Course Outcome: "CO5" / Cognitive Level: "Application"

- B) With neat diagrams, explain (5)
 (I). Satellite-cable television
 (II). Direct-to-Home satellite television

Course Outcome: "CO5" / Cognitive Level: "Application"

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Event Name	Subject
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058-END SEMSTER-ODD-2025 ECE_3126:Satellite
Communication

#	Course Outcome	Description	Marks
1	1	CO 1	7
2	2	CO 2	13
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 ECE_3126:Satellite
Communication

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	2	5(10%)
3	Application	10	45(90%)
4	Analysis	0	0(0%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		12	50 (100%)